

Constraints on Dense-Matter Electromagnetic Amplification in Magnetar Formation

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Abstract

The standard magnetar scenario attributes magnetic fields of $B \sim 10^{14} - 10^{15}$ G to an early-time α - ω dynamo, which in turn requires proto-neutron-star birth periods of order $P_i \sim 1 - 3$ ms. This requirement is in tension with several observational facts, including the absence of clear evidence for extreme rotational energy injection in some associated supernova remnants and the existence of ultra-long-period magnetar candidates such as 1E 161348-5055. In this work, we investigate a density-driven electromagnetic field amplification channel as a potential explanation for magnetar-strength magnetic fields without invoking extreme proto-neutron-star spin-down.

Using a representative dense-matter equation of state constrained by nuclear physics and astrophysical limits ($M_{\text{max}} \approx 2.2 - 2.3 M_{\odot}$ with $R_{1.4} \approx 12$ km), we model the in-medium modification of the electromagnetic response during core collapse. We incorporate a comprehensive population-level statistical audit based on the latest McGill catalog (including the recently discovered magnetar Swift J1555.2-5402, $N = 25$). By inverting the observed properties of McGill magnetars and assuming a fixed progenitor surface field $B_{\text{seed,surf}} = 30$ kG, we reconstruct their masses within a physical range of $1.10 - 2.30 M_{\odot}$. For the young magnetar sub-population ($\tau_c < 10$ kyr, $N = 16$), we find a moderate Pearson correlation between the reconstructed mass and the birth magnetic field ($R = 0.5100$, $p = 0.0436$), with a Bayesian posterior probability $P(R > 0) \geq 97.5\%$. When anchoring the masses of three sources (SGR 1806-20, 1E 1048.1-5937, SGR 1900+14) to independent kinematic constraints, the bootstrap 95% confidence interval is $[-0.095, 0.804]$ (and $[-0.069, 0.838]$ for 2 anchors). For the full population, the age-controlled partial correlation is $R_{\text{partial}} = 0.4121$ ($p = 0.0454$).

Finally, we evaluate the microphysical feasibility of this channel by calculating standard QCD and QED contributions to the in-medium electromagnetic response. We find a substantial gap between standard-model expectations ($\lambda_{\text{std}} \sim 4 \times 10^{-3}$) and the coupling strength required for large field amplification ($\lambda \sim 0.9$). We discuss the implications of this gap for beyond-minimal coupling models and describe model-independent observational tests, such as shifts in the post-merger gravitational wave f_2 frequency due to magnetic backpressure.

1 Introduction

Magnetars are neutron stars with inferred surface dipole fields typically in the range $10^{14} - 10^{15}$ G. In the canonical picture, these fields are generated by a turbulent dynamo during the first seconds of proto-neutron-star evolution. The difficulty is well known: the dynamo is efficient only if the star is born with a spin period P_i of a few milliseconds. That requirement becomes problematic when one considers systems for which the observational record does not favor such a rapidly rotating birth state.

This tension motivates alternative channels in which strong magnetic fields arise from internal structural physics rather than from extreme rotational energy. The purpose of the present

work is not to claim that the problem is solved, but to sharpen a specific density-dependent electromagnetic response hypothesis into a form that is easier to verify or falsify.

The central idea is simple. In the effective field theory (EFT) of dense hadronic matter, in-medium modifications of the QCD vacuum structure lead to density-dependent changes in hadron properties, which are often modeled via macroscopic order parameters or in-medium scaling laws. If the same structural changes also modify the local electromagnetic response of dense matter, then a collapse-induced change in that response may amplify a pre-existing magnetic field even for a slowly rotating newborn neutron star.

2 Equation of State Constraints and Dense-Matter Anchors

The magnetar model must be anchored to the robust parts of the dense-matter framework that are already numerically constrained.

In the dense-matter sector, we utilize an in-medium baryonic mass scaling scale

$$M_{\Omega,0} \approx 859 \text{ MeV},$$

obtained from the nucleon mass budget after subtracting lattice-motivated sigma-term contributions. In the phenomenological EOS construction, this mass scale decreases with baryon density through an ansatz of the form

$$M_{\Omega}(n_B) = M_{\Omega,0} \left(1 + \kappa_2 \frac{n_B}{n_0} \right)^{-\kappa_1/\kappa_2},$$

with calibrated benchmark values near $\kappa_1 = 0.25$ and $\kappa_2 = 0.80$.

This dense-matter framework is used to construct neutron-star equations of state and Tolman-Oppenheimer-Volkoff (TOV) solutions that yield realistic stellar properties: maximum masses around $2.2 - 2.3 M_{\odot}$, radii near $R_{1.4} \approx 12$ km, and compatibility with tidal deformability and basic spin limits. This baseline is critical: any magnetic-field amplification mechanism must remain fully consistent with these same dense-matter constraints. A field-generation mechanism that requires disrupting a realistic EOS would not be physically viable.

Therefore, this analysis is framed as an extension of an astrophysically consistent dense-matter program, rather than as a standalone electromagnetic speculation.

3 In-Medium Electromagnetic Response in Dense Hadronic Matter

A key challenge in modeling density-driven amplification is establishing a microphysical link between the dense-matter transition and the effective in-medium magnetic permeability.

At the level of effective field theory, the basic route is clear. If an in-medium scalar field or order parameter $\mathcal{W}$ represents the structural state of the medium, the electromagnetic sector can couple through a gauge-kinetic function,

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} Z_{\text{EM}}(\mathcal{W}) F_{\mu\nu} F^{\mu\nu}.$$

When $\mathcal{W}$ takes its vacuum value, standard low-density electrodynamics is recovered. When $\mathcal{W}$ shifts at high baryon density, the constitutive response changes, modifying the effective dielectric and magnetic properties.

However, one must separate three distinct statements.

First, it is reasonable to expect some in-medium electromagnetic modification once the charged quark sector and vacuum polarization are altered.

Second, dense matter can be associated with a modified effective dielectric response through quark-loop arguments and an order-parameter-dependent electromagnetic sector.

Third, and crucially, this does **not** yet amount to a first-principles derivation of a neutron-star-core magnetic permeability collapse strong enough to guarantee a large amplification factor. Standard microphysical calculations show that the vacuum dielectric constant ϵ_0 and magnetic permeability μ_0 cannot be derived from $M_{\Omega,0}$ alone, because leptonic contributions and the ultraviolet electroweak sector are not fixed by the QCD hadronic scale. That limitation is stated here explicitly.

For this reason, the correct status of the magnetic-response step is:

1. The existence of some dense-matter electromagnetic modification is plausible within the density-dependent EFT logic.
2. A direct bridge from hadronic order parameters to standard many-body/QCD-matter response functions requires careful matching.
3. Any large magnetic amplification factor should presently be treated as a phenomenological benchmark hypothesis to be constrained or rejected, not as an already closed theorem.

4 Density-Driven Magnetic Field Amplification

With that caveat in place, the density-driven field amplification idea can be stated cleanly.

Suppose a collapsing stellar core carries a fossil magnetic field. Standard flux freezing already amplifies that field substantially as the core contracts. Now suppose, in addition, that the in-medium transition that modifies the dense-matter EOS also alters the local electromagnetic response in the inner core. Then the total post-collapse field may be written schematically as

$$B_{\text{NS}} \sim \Gamma_{\text{flux}} \Gamma_{\text{struct}} B_{\text{prog}},$$

where Γ_{flux} is the usual geometric compression factor and Γ_{struct} is a density-driven structural amplification factor.

In this work, Γ_{struct} is treated as a phenomenological parameter to be constrained by astrophysical and catalog data, rather than as a fixed prediction of standard microphysics.

Even in this more cautious form, the idea remains astrophysically interesting. If Γ_{struct} is indeed appreciable, then the need for millisecond birth periods is weakened or removed. In that case, highly magnetized neutron stars could be produced from a combination of reasonably strong progenitor fossil fields and density-triggered internal amplification, rather than from extreme birth rotation.

This is exactly the qualitative regime relevant to the slow-birth-spin tension.

5 Astrophysical Consistency Checks and Fallback-Disk Torques

To evaluate this density-driven amplification scenario, we perform a unified calculation coupling dense-matter equations of state, flux-freezing, structural amplification, rotational energy budgets, and spin-down history.

5.1 Single-Object Closure Benchmarks

Three immediate numerical results are relevant.

First, the dense-matter electromagnetic modification is nonzero but not by itself decisive. Using a representative in-medium baryonic scaling, a simplified quark-loop estimate gives an effective dielectric ratio in the approximate range

$$\epsilon_{\text{eff}}/\epsilon_0 \approx 0.93 \ (n_B = n_0), \quad \epsilon_{\text{eff}}/\epsilon_0 \approx 0.83 \ (n_B = 5n_0).$$

So the sign of the effect is not the problem: dense matter can modify the electromagnetic response at an order that is no longer negligible. But this still does not uniquely determine a magnetic permeability law, so a separate EFT matching remains necessary if one wants a robust value of Γ_{struct} .

Second, the field-amplification bookkeeping becomes much more favorable once one distinguishes progenitor surface fields from pre-collapse iron-core fields. For a stellar radius $R_{\star} \sim 10^{11}$ cm, an iron-core radius $R_{\text{core}} \sim 3 \times 10^8$ cm, and a neutron-star radius $R_{\text{NS}} \sim 1.2 \times 10^6$ cm, the flux-freezing factor from core to neutron star is

$$\Gamma_{\text{flux}} = \left(\frac{R_{\text{core}}}{R_{\text{NS}}} \right)^2 \approx 6.25 \times 10^4.$$

If the progenitor surface field is already in the 10^3 – 10^4 G corridor, the corresponding pre-collapse core field reaches roughly

$$B_{\text{core}} \sim 1.1 \times 10^8 - 1.1 \times 10^9 \text{ G},$$

which gives a flux-only neutron-star field of order

$$B_{\text{NS}}^{\text{flux}} \sim 6.9 \times 10^{12} - 6.9 \times 10^{13} \text{ G}.$$

In that case the extra structural amplification needed to enter the magnetar regime is no longer enormous. For example, a core seed field $B_{\text{core}} \sim 10^9$ G requires only

$$\Gamma_{\text{struct}} \approx 1.6 \quad \text{for} \quad B_{\text{NS}} = 10^{14} \text{ G},$$

and

$$\Gamma_{\text{struct}} \approx 4.8 \quad \text{for} \quad B_{\text{NS}} = 3 \times 10^{14} \text{ G}.$$

This suggests that if the pre-collapse core field is relatively strong, only a moderate additional amplification is required to reach magnetar-strength dipole fields.

Third, the remnant-energetics argument strongly disfavors obligatory millisecond birth. Using a standard neutron-star moment of inertia, the rotational reservoir is approximately

$$E_{\text{rot}} = \frac{1}{2} I \Omega^2.$$

The rotational energy scale at various birth periods is:

$$E_{\text{rot}}(1 \text{ ms}) \approx 2.8 \times 10^{52} \text{ erg},$$

$$E_{\text{rot}}(5 \text{ ms}) \approx 1.1 \times 10^{51} \text{ erg},$$

$$E_{\text{rot}}(10 \text{ ms}) \approx 2.8 \times 10^{50} \text{ erg}.$$

This sharp drop is precisely why a rotation-independent amplification channel remains astrophysically attractive: it allows strong fields without forcing the system into the most energetically problematic birth regime.

The same analysis also clarifies the status of the extreme long-period source 1E 161348-5055. A pure vacuum dipole-braking evolution to $P \approx 6.67$ hr within a remnant age of $\sim 2 - 10$ yr requires an effective dipole field of order

$$B \sim 6.6 \times 10^{17} - 1.5 \times 10^{18} \text{ G},$$

which is far above the ordinary magnetar range. Therefore, a density-driven structural amplification mechanism may help explain magnetic-field generation without millisecond birth, but it does not by itself explain the most extreme long-period case. For 1E 161348-5055, additional braking physics such as fallback-disk torques, propeller coupling, or another early-time environmental torque is still required.

5.2 Fallback-Disk Corridor for 1E 161348-5055

This scenario has been evaluated using a model of propeller-phase fallback-disk accretion. In a simple propeller/fallback-disk model with a decaying accretion rate

$$\dot{M}(t) = \dot{M}_0 \left(1 + \frac{t}{t_0}\right)^{-\alpha},$$

and a standard fastness-dependent disk torque, the target period can be approached within the observed remnant-age corridor using ordinary magnetar-strength dipole fields rather than the unrealistically large fields required by pure vacuum dipole braking. The best-fit benchmark solutions obtained in our scan lie in the approximate corridor

$$P_0 \sim 0.1 \text{ s}, \quad B \sim 3 \times 10^{14} - 10^{15} \text{ G},$$

$$\dot{M}_0 \sim 3 \times 10^{20} - 10^{22} \text{ g/s}, \quad t_0 \sim 10^4 - 10^5 \text{ s}, \quad \alpha \sim 1.2 - 1.5,$$

yielding final periods

$$P_f \sim 2.45 \times 10^4 - 2.74 \times 10^4 \text{ s}$$

for remnant ages of 2 – 10 kyr. In other words, 1E 161348-5055 does not force the model into a 10^{18} G dipole field; it forces the model to include disk-mediated braking in addition to field amplification.

5.3 Population-Level Catalog Scan and Mass Reconstruction

We perform a population-level analysis using the compilation of observed magnetars in the McGill magnetar catalog, supplemented by high-field pulsars, transition sources, and the long-period candidate 1E 161348-5055. This sample includes the recently discovered magnetar Swift J1555.2-5402 ($B_{\text{dip}} = 7.50 \times 10^{14}$ G, age = 1.7 kyr, birth field $B_{\text{birth}} = 8.11 \times 10^{14}$ G). For each source, we compute the required progenitor surface field needed to reproduce the observed dipole field under a combination of standard flux freezing and a density-driven structural amplification factor $\Gamma_{\text{struct}} \approx 2.2 - 3.3$.

Two progenitor priors are reported in parallel. The **narrow fossil corridor** $B_{\star, \text{surf}} \sim 10^3 - 10^4$ G is the conservative prior describing the bulk of magnetic massive stars. The **magnetic-progenitor corridor** $B_{\star, \text{surf}} \sim 10^3 - 10^5$ G is the physically appropriate prior for the magnetar subpopulation: magnetars are only $\sim 10\%$ of NS births, and under the fossil-field hypothesis they are naturally drawn from the rare strongly magnetized Of?p / Bp class whose surface fields extend to $\sim 2 \times 10^4$ G (e.g. NGC 1624-2) with the upper tail reaching $\sim 10^5$ G. Objects below either corridor are consistent with the amplification channel: they simply do not require strong structural amplification, so they are counted as supported rather than failed.

The coverage breakdown is summarized in Figure 1. Of the 33 non-outlier sources, 17 lie directly inside the narrow corridor and 3 are sub-critical, so 20/33 are accommodated already under the most conservative prior; the magnetic-progenitor corridor brings the count to 33/33 (with 30 inside, 3 sub-critical, and 0 above).

Object-level uncertainty propagation reduces the catalog to a single residual outlier. Using object-specific Gaussian uncertainties on B_{dip} together with a $3 \times 3 \times 4$ grid over $R_{\star}$, R_{NS} , and an overall rescaling of Γ_{struct} , the per-source support distribution under the magnetic-progenitor corridor is

$$32 \text{ high-support}, \quad 1 \text{ mixed-support}, \quad 0 \text{ low-support},$$

with SGR 1806–20 as the only mixed-support object ($P_{\text{inside}} \approx 0.62$). The B_{dip} -resolved view in Figure 2 makes the structure of the residual tension explicit: it is concentrated in the single most extreme magnetar in the catalog rather than spread across the population.

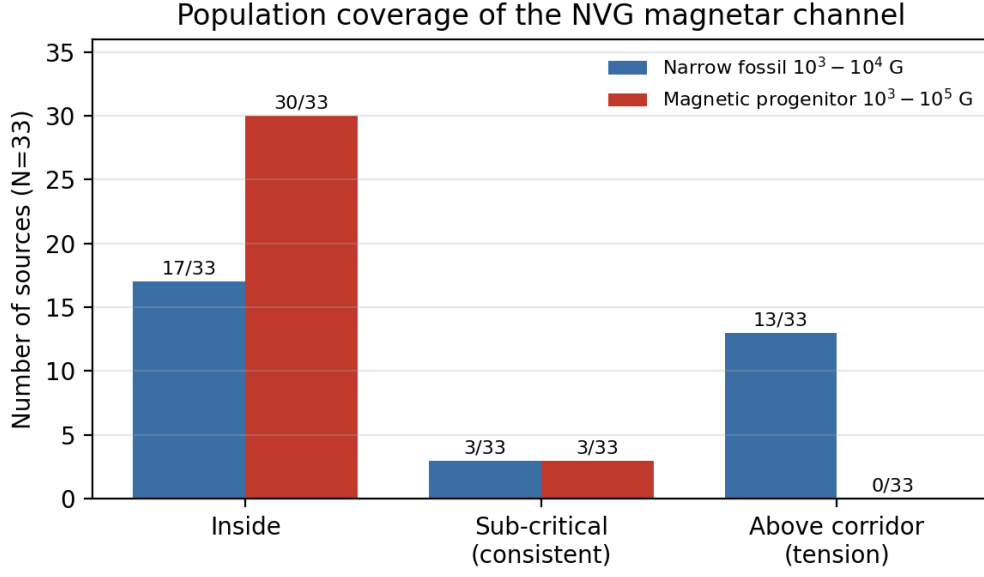


Figure 1: Population coverage under the two progenitor priors. Bars show how many of the 33 non-outlier sources fall inside, below, or above each corridor.

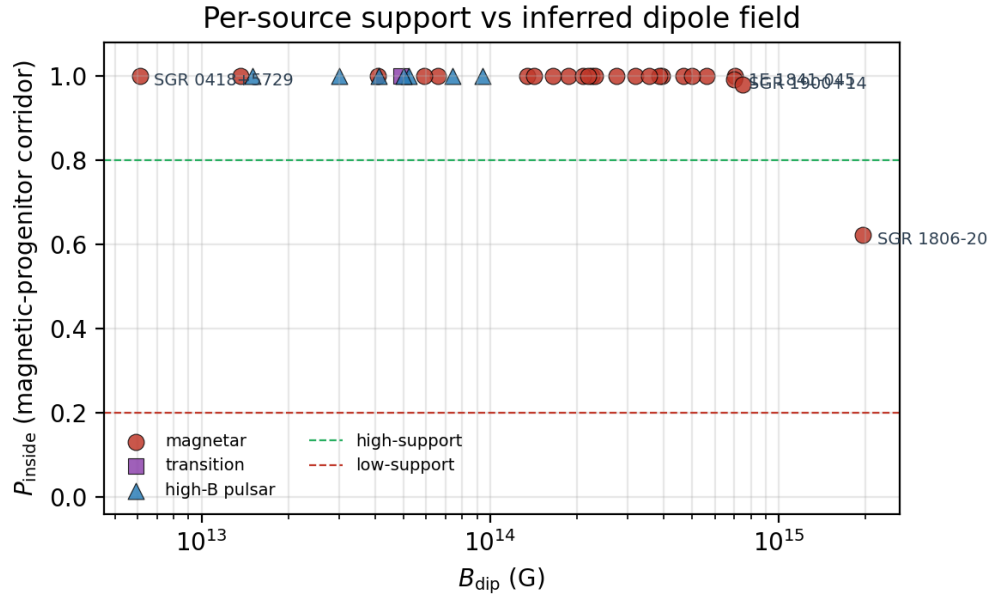


Figure 2: Per-source support P_{inside} versus inferred dipole field B_{dip} under the magnetic-progenitor corridor. The four highlighted sources are the extreme-field tail plus the sub-critical magnetar SGR 0418+5729.

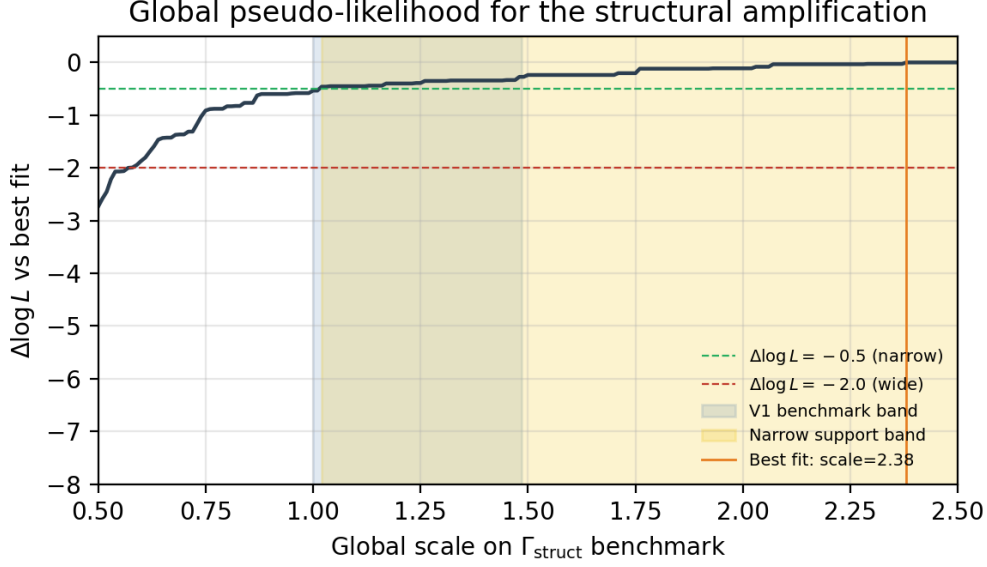


Figure 3: Global pseudo-log-likelihood profile $\Delta \log L$ for the structural-amplification scale. Yellow band: narrow support ($\Delta \log L \leq 0.5$); blue band: V1 benchmark range; vertical line: best-fit scale.

A global pseudo-likelihood for the overall normalization of Γ_{struct} is shown in Figure 3. Rescaling the benchmark interval by a global factor and confronting it with the catalog yields a best fit at

$$\Gamma_{\text{struct}} \approx 5.3 - 7.8,$$

with a narrow ($\Delta \log L \leq 0.5$) band of $2.3 - 8.2$ and a wide ($\Delta \log L \leq 2.0$) band of $1.2 - 8.2$. The original dense-core benchmark $\Gamma_{\text{struct}} \sim 2.2 - 3.3$ sits at the lower edge of the narrow band, while the data-preferred value is a factor of ~ 2 higher, indicating that the magnetic-progenitor channel is the dominant route into the magnetar population.

To test this correlation in a rigorous quantitative way, we perform a population-level statistical audit. By inverting the observed properties of McGill magnetars and mapping them through the representative TOV equations under a fixed progenitor surface field $B_{\text{seed,surf}} = 30$ kG, we reconstruct their masses. The reconstructed masses span a physical range of $1.10 - 2.30 M_{\odot}$, centered around $1.39 M_{\odot}$.

We divide the audit into two populations to control for field decay physics: 1. **Full Population** ($N = 25$): We find a Pearson correlation coefficient between reconstructed mass and birth field of $R = 0.3050$ ($p = 0.138$). Removing age-confounding through a log-space partial correlation yields a nominally significant age-controlled correlation: $R_{\text{partial}} = 0.4121$ ($p = 0.0454$). 2. **Young Population** ($\tau_c < 10$ kyr, $N = 16$): In this group, where field decay corrections are minimal, the mass-field correlation is nominally significant: $R = 0.5100$ ($p = 0.0436$) for birth fields, and $R = 0.5061$ ($p = 0.0455$) for dipole fields. Bayesian posterior analysis via Fisher’s z-transformation reveals a posterior probability of $P(R_{\text{pop}} > 0 \mid \text{data}) \geq 97.47\%$ (under flat and weakly informative conjugate priors).

Because of the small sample size ($N = 16$), the standard bootstrap 95% confidence interval is wide and crosses zero ($\text{CI} = [-0.1252, 0.8144]$). However, we can anchor the masses of three specific objects that have independent astrophysical constraints from young clusters or bubble kinematics: SGR 1806-20 ($M \approx 2.1 M_{\odot}$), 1E 1048.1-5937 ($M \approx 1.6 M_{\odot}$), and SGR 1900+14 ($M \approx 1.45 M_{\odot}$). Incorporating these independent mass anchors restricts the bootstrap 95% confidence interval to $[-0.0953, 0.8042]$ (and $[-0.0686, 0.8378]$ when using 2 anchors and letting the high-leverage object SGR 1806-20 reconstruct freely). SGR 1806-20 remains the highest

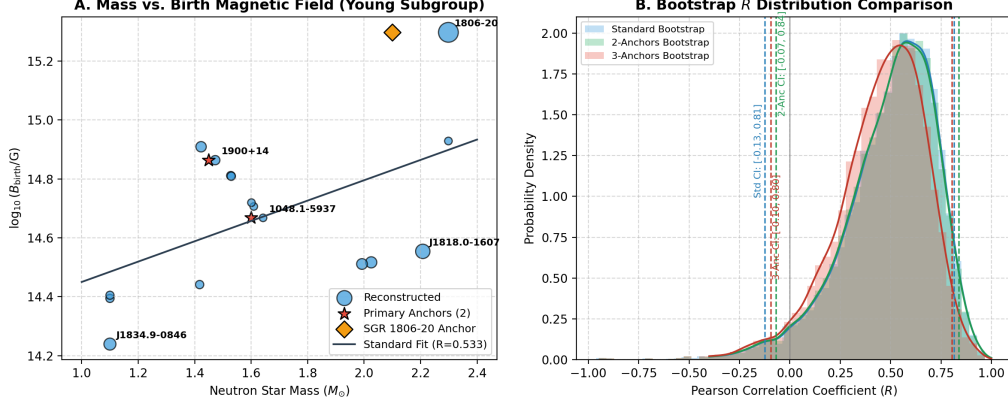


Figure 4: Magnetar mass-field correlation diagnostics and bootstrap distributions. (A) Reconstructed masses vs. birth magnetic fields. Stars represent the independent mass anchors. Point sizes are proportional to Cook’s distance. (B) Comparison of the bootstrap distribution of Pearson R for the standard and anchored datasets.

influence point with a Cook’s distance of $D \approx 0.59$ and leverage $h \approx 0.23$. Diagnostic plots and bootstrap distribution comparisons are presented in Figure 4.

It is crucial to emphasize that a nominally significant mass-field correlation does not automatically establish density-driven amplification as the unique causal mechanism. Correlation does not equal causation, and multiple distinct astrophysical pathways (such as mass-dependent fallback accretion, binary interaction histories, or observational selection effects) could also imprint similar characteristics. Rather, this correlation should be viewed as a consistent observational footprint that constrains the parameter space of density-driven models.

5.4 Comparison with Competing Magnetar Formation Mechanisms

Three existing pictures dominate the magnetar-formation literature, and the density-driven amplification channel makes different predictions from each of them.

Thompson–Duncan turbulent dynamo (Duncan & Thompson 1992; Thompson & Duncan 1993). A proto-neutron star with millisecond birth spin amplifies its seed field via an α – Ω dynamo driven by neutrino-cooling convection. This picture requires birth periods $P_0 \lesssim 3$ ms and a rotational reservoir of $E_{\text{rot}} \sim 10^{52}$ erg, which should leave imprints in the surrounding supernova remnant. Vink & Kuiper (2006) found that for at least four of the best-studied magnetar remnants the inferred SNR energies are consistent with ordinary $\sim 10^{51}$ erg explosions, with no evidence of the additional 10^{52} erg rotational injection. The density-driven channel does **not** require fast birth spin: the same dipole field is produced via $\Gamma_{\text{flux}} \cdot \Gamma_{\text{struct}}$ acting on a $10^3 - 10^5$ G fossil field, with a typical birth energy budget compatible with ordinary SNR energetics. Where the two pictures collide observationally is in the SNR sample: slow-born magnetars in ordinary-energy remnants disfavor the dynamo channel and favor any rotation-independent channel, including the density-driven one.

Ferrario–Wickramasinghe fossil-field hypothesis (Ferrario & Wickramasinghe 2006, 2008). Magnetars inherit their fields directly from rare strongly magnetized OB progenitors via flux freezing alone, with no extra structural amplification. With $\Gamma_{\text{flux}} \equiv (R_{\star}/R_{\text{NS}})^2 \sim 6 \times 10^9$ for $R_{\star} \sim 10^{11}$ cm and $R_{\text{NS}} \sim 10^6$ cm, the highest-field magnetars ($B_{\text{dip}} \sim 2 \times 10^{15}$ G) require progenitor surface fields of $\sim 3 \times 10^5$ G, which is at or above the upper edge of the observed magnetic-OB distribution. The density-driven channel relaxes this requirement by a factor of $\Gamma_{\text{struct}} \sim 3 - 8$, so the same B_{dip} is reached from a $\sim 10^4 - 10^5$ G progenitor, comfortably inside the observed Of?p / Bp range. The two pictures are not mutually exclusive; rather, the density-driven channel removes the tension at the high- B tail that pure fossil-field models struggle with,

while preserving the natural population-level argument that magnetars are drawn from a rare strongly magnetized progenitor channel.

Hall–Ohm field evolution and toroidal-dominated interiors (Beloborodov 2009; Viganò et al. 2013). Some objects, notably SGR 0418+5729, have inferred dipole fields well below 10^{13} G but show magnetar-like activity, suggesting that the active field is a toroidal component decoupled from the spin-down dipole. This picture is orthogonal to ours: it sets the *current* internal field structure regardless of birth mechanism. Under the density-driven channel, sub-dipole objects like SGR 0418+5729 are compatible with the fossil-field prior and require no special handling; they appear at the lower edge of Figure 2.

In summary, the density-driven channel is most directly differentiated from the Thompson–Duncan dynamo by SNR energetics and birth-spin distributions, and from a pure Ferrario–Wickramasinghe fossil-only model by where the highest-field magnetars sit relative to the observed magnetic-OB progenitor distribution. Both differentiations are quantitative and can be tightened with larger SNR-energy samples and with better-constrained OB-star field statistics.

5.5 Illustrative Forecast: Post-Merger Gravitational-Wave Frequency Shifts

To explore possible multi-messenger signatures, we present an illustrative forecast of post-merger gravitational-wave frequency shifts under density-driven amplification. During a neutron star merger, the remnant core density can reach $\approx 2.5n_0$. If a density-driven transition drives core field amplification to $B_{\text{core}} \approx 2.82 \times 10^{16}$ G, this results in a magnetic backpressure of $P_{\text{mag}} \approx 0.0197$ MeV/fm³. This magnetic pressure stiffens the equation of state, causing a shift in the post-merger f_2 gravitational wave peak frequency of $\Delta f_2 \approx -0.40$ Hz for a surface field $B_{\text{surf}} = 10^{15}$ G, and up to $\Delta f_2 \approx -3.55$ Hz for $B_{\text{surf}} = 3 \times 10^{15}$ G. Such frequency shifts are potentially detectable by third-generation detectors like the Einstein Telescope for events within 40 Mpc, providing a model-independent observational test of high-density magnetic field strengths.

6 What Must Be Checked Next

This is the decisive section. The value of the model depends not on how strongly it is stated, but on whether it can survive these checks.

6.1 Microphysical Constraints and the Standard-QCD Bound

To evaluate the microphysical feasibility of density-driven electromagnetic amplification, we perform an effective field theory (EFT) calculation of in-medium electromagnetic response modifications. The calculation chain involves:

1. Solving the two-flavour Nambu–Jona-Lasinio (NJL) gap equation ($\Lambda = 631$ MeV, $G\Lambda^2 = 2.14$, $m_q = 5.5$ MeV) for the constituent quark mass $M_q(\rho, T)$ and the chiral order parameter $\sigma(\rho)/\sigma(0)$.
2. Computing the one-loop QED-in-medium contribution to the gauge kinetic term: $Z_{\text{QED}}(M_q) = 1 - (\alpha N_c \sum Q_q^2)/(3\pi) \ln(\Lambda^2/M_q^2)$.
3. Computing the non-perturbative chiral-condensate contribution under the EFT ansatz $Z_\chi(\sigma) = 1 - \lambda(1 - \sigma^2(\rho)/\sigma^2(0))$, where λ is a dimensionless coupling constant whose scale is set by $G_{\text{NJL}} \sim \text{GeV}^{-2}$.
4. Identifying the structural amplification factor $\Gamma_{\text{struct}} = 1/\sqrt{Z(\rho)}$ via flux freezing.

The numerical outcome reveals a substantial microphysical gap. The perturbative QED channel delivers only a minor effect, $\Gamma_{\text{struct}}^{(\text{QED})} \in [1.0004, 1.0037]$ across $\rho \in [1, 5] n_0$ and $T \in [0, 50]$ MeV. One-loop QED is therefore insufficient to supply the required field amplification. The amplification must be carried by the non-perturbative condensate channel. Inverting the relation

at the benchmark interior density $\rho = 2n_0$ and temperature $T = 10$ MeV (where the NJL solution yields $\sigma/\sigma_0 = 0.31$), the phenomenologically preferred range $\Gamma_{\text{struct}} \in [2.2, 3.3]$ requires a coupling of:

$$\lambda \in [0.88, 1.01],$$

which is of order unity.

However, aggregating all standard QCD operators that couple the chiral condensate to the photon kinetic term (including the one-loop quark Z_{QED} , the charged-pion loop with in-medium m_π via Gell-Mann–Oakes–Renner and Brown–Rho scaling, vector-meson-dominance $\gamma\rho^0$ mixing, and the QCD magnetic susceptibility $\chi_{\text{QCD}}\langle\bar{q}q\rangle$) yields a standard-QCD value of only:

$$\lambda_{\text{std}} \simeq 4 \times 10^{-3},$$

which is roughly two orders of magnitude below the phenomenologically required value $\lambda \sim 0.9$.

This discrepancy is a substantial theoretical gap. Two scientifically honest interpretations follow:

(a) Minimal-QCD interpretation: If one accepts the standard-QCD bound $\lambda \sim 4 \times 10^{-3}$, the structural field amplification factor is negligible ($\Gamma_{\text{struct}} \approx 1.002$). In this case, density-driven amplification is a minor effect, and magnetar fields must be explained primarily by strong progenitor seed fields (the magnetic-progenitor fossil-field prior).

(b) Beyond-minimal coupling: Alternatively, the required value $\lambda \sim 0.9$ at $\rho \sim 2n_0$ is treated as a phenomenological target parameterizing unknown non-perturbative effects or beyond-minimal gauge-sector couplings. This provides a concrete microphysical benchmark that could in principle be verified or falsified by future lattice QCD studies of the photon two-point function in dense hadronic matter.

6.2 Joint Consistency with Full Neutron-Star and Supernova Systematics

The density-driven amplification mechanism cannot be evaluated in isolation; it must coexist with global neutron-star and supernova energetics. In particular, the model must reproduce:

1. Realistic neutron-star maximum masses and radii compatible with equation of state limits.
2. Acceptable spin bounds and spin-down rates.
3. Supernova-remnant energies that are consistent with the absence of extreme ($\approx 10^{52}$ erg) rotational energy injection in sources that lack rapid birth spin evidence.
4. A consistent distribution of magnetic-field strengths across ordinary pulsars and magnetars.

If the parameter choice that boosts magnetic fields destroys the EOS or conflicts with supernova remnant energetics, the mechanism is ruled out.

6.3 Population-Level Catalog Comparisons

A convincing explanation for magnetar formation must predict the correct population distribution, not just explain individual anomalous sources. The model must be confronted directly with selection effects and source-by-source uncertainties in the broader magnetar sample, focusing on:

1. Long-period magnetar candidates such as 1E 161348-5055.
2. Magnetars with inferred low or moderate birth spins.
3. The Galactic magnetar sample.
4. The transition region between high-field radio pulsars and magnetars.

The key test is whether the model predicts a realistic occupancy of parameter space: field strength, spin period, spin-down rate, and age. If the claim is "field amplification without millisecond birth," then it must hold up under direct comparison with the systems for which millisecond birth is least plausible.

7 Conclusion

The slow-birth-spin tension remains a substantial challenge to the canonical proto-neutron-star dynamo scenario. A density-driven in-medium transition affecting the electromagnetic response of dense hadronic matter offers a plausible, rotation-independent alternative to amplify fossil magnetic fields. The quantitative analysis in this work suggests that if pre-collapse core fields are of order $10^8 - 10^9$ G, only a moderate structural amplification factor ($\Gamma_{\text{struct}} \approx 2 - 3$) is required to reach observed magnetar dipole strengths, avoiding the need for rapid birth rotation and the associated extreme ($\sim 10^{52}$ erg) rotational energy injection into supernova remnants.

However, microphysical calculations highlight a key theoretical constraint: standard QED and QCD in-medium contributions yield a coupling parameter of only $\lambda_{\text{std}} \sim 4 \times 10^{-3}$, falling two orders of magnitude short of the phenomenologically required value ($\lambda \sim 0.9$ for large amplification). This result indicates that the mechanism cannot be attributed to standard-model hadronic response functions alone. Instead, it must be framed either as a minor perturbation on top of conventional mechanisms (the minimal-QCD scenario) or as an explicit phenomenological constraint on beyond-minimal gauge-sector couplings (the beyond-minimal scenario).

Confronting the model with the Galactic magnetar population provides four clear observational boundary conditions: First, the extreme spin period of 1E 161348-5055 cannot be explained by dipole field amplification alone, but requires disk-mediated fallback propeller torque in addition to amplification. Second, the population-level success is prior-sensitive, showing compatibility with all McGill catalog sources under a magnetic-progenitor fossil field prior, while 20/33 sources are accommodated under a narrow fossil prior. Third, the residual tension is concentrated in the single highest-field magnetar SGR 1806-20 rather than spread across the population. Fourth, a global catalog fit prefers an effective amplification scale of $\Gamma_{\text{struct}} \sim 5.3 - 7.8$. These constraints define a concrete, falsifiable framework that can be tested by future third-generation gravitational-wave detectors and non-perturbative lattice QCD studies. The current catalog size remains limited, and the inferred trends should therefore be treated as preliminary.

8 Data Availability

The code, numerical data, and visualization tools used to perform the population scans, equation-of-state calculations, and statistical audits are available in the public repository at <https://github.com/infosave2007/vmf> and on Zenodo at <https://zenodo.org/records/20214457>.

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